

FIFA World Cup Analysis

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Subject Name: Analytics Programming

GitHub Repository:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission>

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Question 1: Data Inspection, Cleaning and Home Penalty Distribution

Rationale

I had to combine the four data sets because the data needed for our analysis did not reside within a single file. The `Matches` data set gave the results of the matches and their scores while the other data sets provided the information on teams, stadiums and tournaments. Combining the four data sets allowed me to pull all the info together in one data frame.

I replaced "?" with NA because while "?" is just a string, NA is recognized by R as missing information. I converted `Result`, `Stage`, `Country` and `ExtraTime` into factors since these are categorical variables. I changed variables `HomePenalty` and `AwayPenalty` into numbers since they reflect the amount of penalty kicks taken.

I used a bar chart for `HomePenalty` as these are discrete whole numbers. This makes it easy to see what penalty kicks had bigger frequencies compared to others.

Load and Join the Data

```
# Read the data CSV files.
matches <- read_csv("../data/Matches.csv")
stadiums <- read_csv("../data/Stadiums.csv")
teams <- read_csv("../data/Teams.csv")
tournaments <- read_csv("../data/Tournaments.csv")

# Change every "?" value to NA.
matches[matches == "?"] <- NA
stadiums[stadiums == "?"] <- NA
teams[teams == "?"] <- NA
tournaments[tournaments == "?"] <- NA

# Join team information for the home team and away team.
df <- matches %>%
  left_join(
    teams %>%
      rename(
        HomeTeamID = TeamID,
        HomeTeamName = TeamName,
        HomeTeamCode = TeamCode
      ),
    by = "HomeTeamID"
  ) %>%
  left_join(
    teams %>%
      rename(
        AwayTeamID = TeamID,
        AwayTeamName = TeamName,
        AwayTeamCode = TeamCode
      ),
    by = "AwayTeamID"
  ) %>%
```

```
left_join(stadiums, by = "StadiumID") %>%
left_join(tournaments, by = "TournamentID")
```

Inspect and Clean the Data

```
# Display the first six rows.
head(df)
```

```
## # A tibble: 6 x 24
##   MatchID TournamentID MatchName Stage MatchDate MatchTime StadiumID HomeTeamID
##   <chr>      <chr>      <chr>    <chr> <chr>      <time>      <chr>      <chr>
## 1 M-1930-- WC-1930      France v~ grou~ 7/13/1930 15:00      S-193      T-28
## 2 M-1930-- WC-1930      United S~ grou~ 7/13/1930 15:00      S-192      T-80
## 3 M-1930-- WC-1930      Yugoslav~ grou~ 7/14/1930 12:45      S-192      T-84
## 4 M-1930-- WC-1930      Romania ~ grou~ 7/14/1930 14:50      S-193      T-59
## 5 M-1930-- WC-1930      Argentin~ grou~ 7/15/1930 16:00      S-192      T-03
## 6 M-1930-- WC-1930      Chile v ~ grou~ 7/16/1930 14:45      S-192      T-13
## # i 16 more variables: AwayTeamID <chr>, HomeTeamScore <dbl>,
## #   AwayTeamScore <dbl>, ExtraTime <dbl>, PenaltyShootout <dbl>,
## #   HomePenalty <chr>, AwayPenalty <chr>, Result <chr>, HomeTeamName <chr>,
## #   HomeTeamCode <chr>, AwayTeamName <chr>, AwayTeamCode <chr>,
## #   StadiumName <chr>, City <chr>, Country <chr>, TournamentName <chr>
```

```
# Check the number of rows and columns.
dim(df)
```

```
## [1] 964 24
```

```
# Check the variable types.
str(df)
```

```
## spc_tbl_ [964 x 24] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ MatchID      : chr [1:964] "M-1930-01" "M-1930-02" "M-1930-03" "M-1930-04" ...
## $ TournamentID : chr [1:964] "WC-1930" "WC-1930" "WC-1930" "WC-1930" ...
## $ MatchName    : chr [1:964] "France v Mexico" "United States v Belgium" "Yugoslavia v Brazil" "R
## $ Stage        : chr [1:964] "group stage" "group stage" "group stage" "group stage" ...
## $ MatchDate    : chr [1:964] "7/13/1930" "7/13/1930" "7/14/1930" "7/14/1930" ...
## $ MatchTime    : 'hms' num [1:964] 15:00:00 15:00:00 12:45:00 14:50:00 ...
## ..- attr(*, "units")= chr "secs"
## $ StadiumID    : chr [1:964] "S-193" "S-192" "S-192" "S-193" ...
## $ HomeTeamID   : chr [1:964] "T-28" "T-80" "T-84" "T-59" ...
## $ AwayTeamID   : chr [1:964] "T-44" "T-06" "T-09" "T-54" ...
## $ HomeTeamScore : num [1:964] 4 3 2 3 1 3 4 3 1 1 ...
## $ AwayTeamScore : num [1:964] 1 0 1 1 0 0 0 0 0 0 ...
## $ ExtraTime    : num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ PenaltyShootout: num [1:964] 0 0 0 0 0 0 0 0 0 0 ...
## $ HomePenalty  : chr [1:964] NA "0" "0" "0" ...
## $ AwayPenalty  : chr [1:964] "0" "0" "0" NA ...
## $ Result       : chr [1:964] "home team win" "home team win" "home team win" "home team win" ...
## $ HomeTeamName  : chr [1:964] "France" "United States" "Yugoslavia" "Romania" ...
```

```
## $ HomeTeamCode : chr [1:964] "FRA" "USA" "YUG" "ROU" ...
## $ AwayTeamName : chr [1:964] "Mexico" "Belgium" "Brazil" "Peru" ...
## $ AwayTeamCode : chr [1:964] "MEX" "BEL" "BRA" "PER" ...
## $ StadiumName : chr [1:964] "Estadio Pocitos" "Estadio Gran Parque Central" "Estadio Gran Parque
## $ City : chr [1:964] "Montevideo" "Montevideo" "Montevideo" "Montevideo" ...
## $ Country : chr [1:964] "Uruguay" "Uruguay" "Uruguay" "Uruguay" ...
## $ TournamentName : chr [1:964] "1930 FIFA World Cup" "1930 FIFA World Cup" "1930 FIFA World Cup" "1
## - attr(*, "spec")=
## .. cols(
## .. MatchID = col_character(),
## .. TournamentID = col_character(),
## .. MatchName = col_character(),
## .. Stage = col_character(),
## .. MatchDate = col_character(),
## .. MatchTime = col_time(format = ""),
## .. StadiumID = col_character(),
## .. HomeTeamID = col_character(),
## .. AwayTeamID = col_character(),
## .. HomeTeamScore = col_double(),
## .. AwayTeamScore = col_double(),
## .. ExtraTime = col_double(),
## .. PenaltyShootout = col_double(),
## .. HomePenalty = col_character(),
## .. AwayPenalty = col_character(),
## .. Result = col_character()
## .. )
## - attr(*, "problems")=<externalptr>
```

Display a summary of the main variables.

```
summary(
  df[c(
    "Result",
    "Stage",
    "Country",
    "ExtraTime",
    "PenaltyShootout",
    "HomePenalty",
    "AwayPenalty"
  )]
)
```

##	Result	Stage	Country	ExtraTime
##	Length:964	Length:964	Length:964	Min. :0.00000
##	Class :character	Class :character	Class :character	1st Qu.:0.00000
##	Mode :character	Mode :character	Mode :character	Median :0.00000
##				Mean :0.07573
##				3rd Qu.:0.00000
##				Max. :1.00000
##	PenaltyShootout	HomePenalty	AwayPenalty	
##	Min. :0.00000	Length:964	Length:964	
##	1st Qu.:0.00000	Class :character	Class :character	
##	Median :0.00000	Mode :character	Mode :character	
##	Mean :0.03631			
##	3rd Qu.:0.00000			

```
## Max. :1.00000
```

```
# Convert categorical variables to factors.
df$Result <- as.factor(df$Result)
df$Stage <- as.factor(df$Stage)
df$Country <- as.factor(df$Country)
df$ExtraTime <- as.factor(df$ExtraTime)

# Convert penalty variables to numeric.
df$HomePenalty <- as.numeric(df$HomePenalty)
df$AwayPenalty <- as.numeric(df$AwayPenalty)

# Replace missing penalty values with zero.
df$HomePenalty[is.na(df$HomePenalty)] <- 0
df$AwayPenalty[is.na(df$AwayPenalty)] <- 0
```

Code Testing

```
# The number of rows should be unchanged after joining.
nrow(matches)
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

```
# Check the classes after conversion.
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$Country)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
class(df$HomePenalty)
```

```
## [1] "numeric"
```

```
class(df$AwayPenalty)
```

```
## [1] "numeric"
```

```
# Check that no missing penalty values remain.
```

```
sum(is.na(df$HomePenalty))
```

```
## [1] 0
```

```
sum(is.na(df$AwayPenalty))
```

```
## [1] 0
```

The count of rows in the `matches` and `df` matrices ought to be identical. This is critical, as joining the other datasets would only increase the number of columns but not the number of match records.

The results of the four categorical variables should be `"factor"`. The outcome for the penalty variables should be `"numeric"`. The results of the two missing value checks should both be zeros. This will allow one to verify that the data has been converted into the appropriate format for analysis.

Results and Interpretation

```
# Keep only matches involving a penalty shootout.
```

```
penalty_matches <- df %>%  
  filter(PenaltyShootout == 1)
```

```
# Create a frequency table for HomePenalty.
```

```
penalty_table <- table(penalty_matches$HomePenalty)  
penalty_table
```

```
##
```

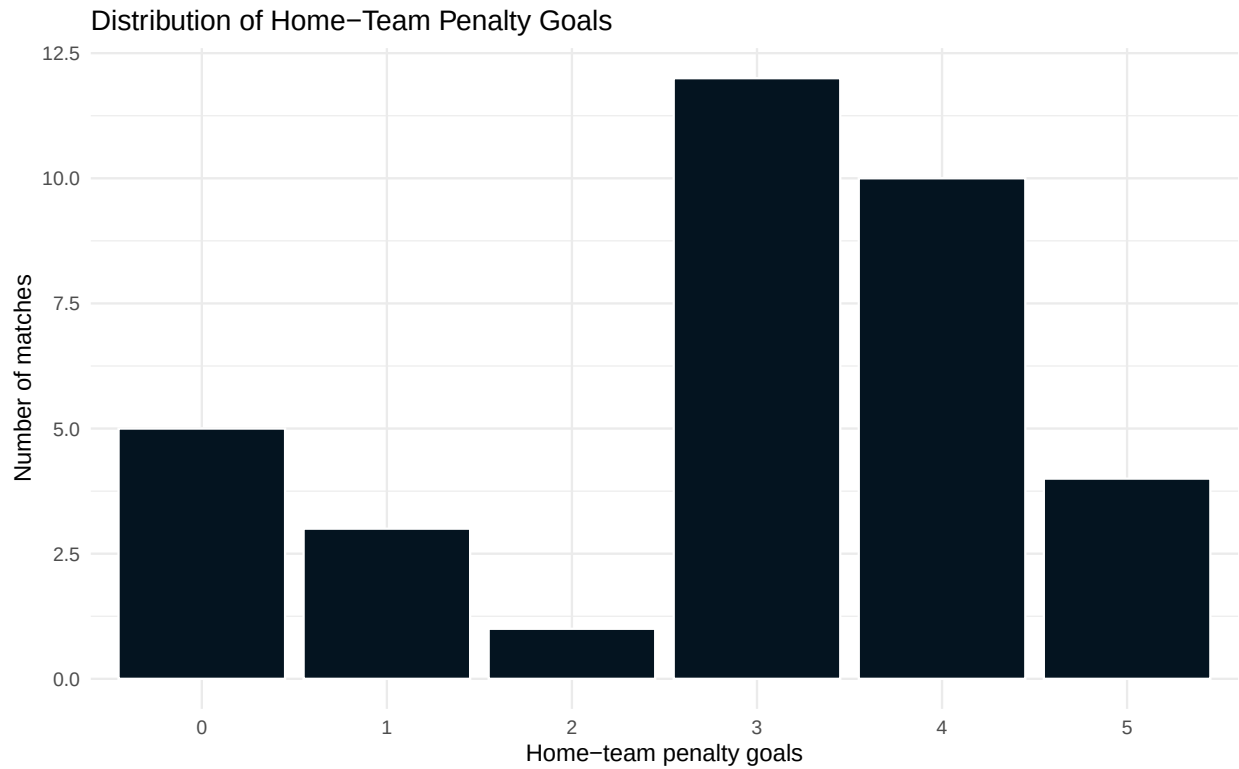
```
## 0  1  2  3  4  5
```

```
## 5  3  1 12 10  4
```

```
# Find the most common HomePenalty value.
```

```
most_common_penalty <- names(penalty_table)[which.max(penalty_table)]  
most_common_count <- max(penalty_table)
```

```
ggplot(penalty_matches, aes(x = factor(HomePenalty))) +  
  geom_bar(  
    fill = darkest_colour,  
    colour = "white"  
  ) +  
  labs(  
    title = "Distribution of Home-Team Penalty Goals",  
    x = "Home-team penalty goals",  
    y = "Number of matches"  
  ) +  
  theme_minimal()
```



Out of a total of 35 matches, 12 matches had home penalty of 3, which was most commonly observed.

According to the bar chart, some penalty scores received more frequent occurrences than others. It can be inferred that as the size of the column increases, the occurrence of that particular score increases. On the contrary, smaller the length of the column; lesser will be the occurrence of that score. So, the distribution is not uniform as heights of the columns are different.

Question 2: Distribution of Home Team Scores

Rationale

The reason I employed histograms is that the `HomeTeamScore` is a numerical variable. In football, scores are expressed as integers so I set a bin width of 1. It implies that each bar represents a different score such as 0, 1, 2 or 3 goals.

I made comparisons of home team scores based on `Stage`, `TournamentName` and `AwayScoreGroup` separately so that encoded information was not overlapping and was easier to read.

In addition, I calculated the mean and median using `aggregate()` function for every group.

Distribution by Competition Stage

```
# Calculate the mean and median HomeTeamScore for each stage.
stage_mean <- aggregate(
  HomeTeamScore ~ Stage,
  mean,
  data = df
)

stage_median <- aggregate(
  HomeTeamScore ~ Stage,
  median,
  data = df
)

stage_mean
```

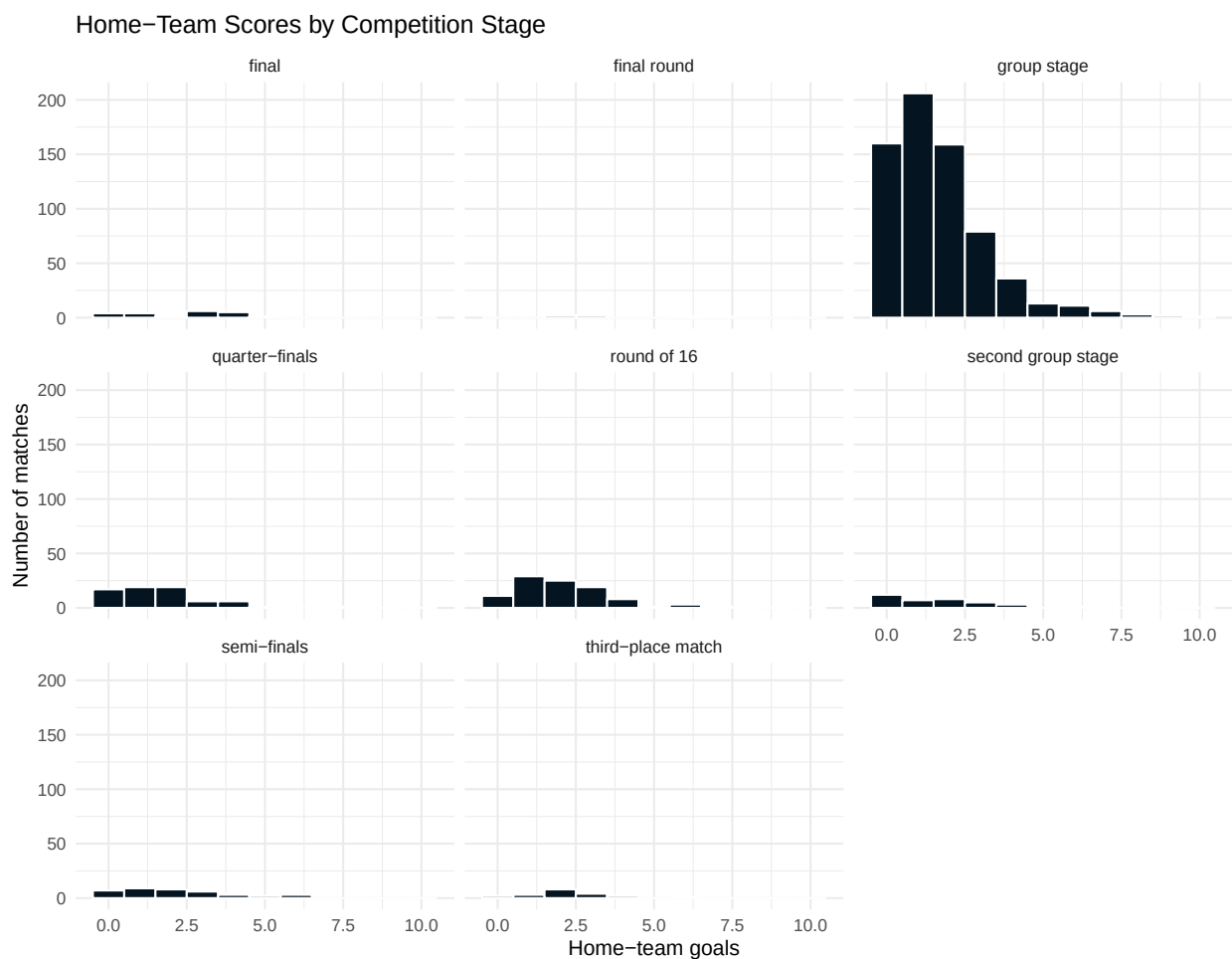
```
##           Stage HomeTeamScore
## 1           final      2.333333
## 2      final round      3.833333
## 3      group stage      1.671598
## 4 quarter-finals      1.700000
## 5    round of 16      2.041237
## 6 second group stage      1.555556
## 7      semi-finals      2.184211
## 8 third-place match      2.250000
```

```
stage_median
```

```
##           Stage HomeTeamScore
## 1           final            3
## 2      final round            3
## 3      group stage            1
## 4 quarter-finals            1
## 5    round of 16             2
## 6 second group stage         1
## 7      semi-finals           2
## 8 third-place match          2
```



```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ Stage) +
  labs(
    title = "Home-Team Scores by Competition Stage",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Distribution by Tournament Name

```
# Calculate the mean and median HomeTeamScore for each tournament.
tournament_mean <- aggregate(
  HomeTeamScore ~ TournamentName,
  mean,
```

```

data = df
)

tournament_median <- aggregate(
  HomeTeamScore ~ TournamentName,
  median,
  data = df
)

tournament_mean

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup      3.277778
## 2  1934 FIFA World Cup      2.823529
## 3  1938 FIFA World Cup      3.388889
## 4  1950 FIFA World Cup      3.136364
## 5  1954 FIFA World Cup      4.192308
## 6  1958 FIFA World Cup      2.514286
## 7  1962 FIFA World Cup      2.156250
## 8  1966 FIFA World Cup      2.156250
## 9  1970 FIFA World Cup      2.218750
## 10 1974 FIFA World Cup      1.342105
## 11 1978 FIFA World Cup      1.894737
## 12 1982 FIFA World Cup      1.846154
## 13 1986 FIFA World Cup      1.384615
## 14 1990 FIFA World Cup      1.230769
## 15 1994 FIFA World Cup      1.596154
## 16 1998 FIFA World Cup      1.531250
## 17 2002 FIFA World Cup      1.390625
## 18 2006 FIFA World Cup      1.343750
## 19 2010 FIFA World Cup      1.187500
## 20 2014 FIFA World Cup      1.265625
## 21 2018 FIFA World Cup      1.421875
## 22 2022 FIFA World Cup      1.578125

```

```

tournament_median

```

```

##      TournamentName HomeTeamScore
## 1  1930 FIFA World Cup          3.0
## 2  1934 FIFA World Cup          3.0
## 3  1938 FIFA World Cup          3.0
## 4  1950 FIFA World Cup          2.5
## 5  1954 FIFA World Cup          4.0
## 6  1958 FIFA World Cup          2.0
## 7  1962 FIFA World Cup          2.0
## 8  1966 FIFA World Cup          2.0
## 9  1970 FIFA World Cup          2.0
## 10 1974 FIFA World Cup          1.0
## 11 1978 FIFA World Cup          2.0
## 12 1982 FIFA World Cup          1.0
## 13 1986 FIFA World Cup          1.0
## 14 1990 FIFA World Cup          1.0
## 15 1994 FIFA World Cup          1.0

```

```
## 16 1998 FIFA World Cup      1.0
## 17 2002 FIFA World Cup      1.0
## 18 2006 FIFA World Cup      1.0
## 19 2010 FIFA World Cup      1.0
## 20 2014 FIFA World Cup      1.0
## 21 2018 FIFA World Cup      1.0
## 22 2022 FIFA World Cup      1.0
```

```
ggplot(df, aes(x = HomeTeamScore)) +
  geom_histogram(
    binwidth = 1,
    fill = darkest_colour,
    colour = "white"
  ) +
  facet_wrap(~ TournamentName) +
  labs(
    title = "Home-Team Scores by Tournament",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```

Home-Team Scores by Tournament



Distribution by Away-Team Score Group

```
# Create an empty variable for the away-team score groups.
df$AwayScoreGroup <- NA
```

```

# Assign each match to one of the three required groups.
df$AwayScoreGroup[df$AwayTeamScore <= 1] <- "0-1 goals"

df$AwayScoreGroup[
  df$AwayTeamScore >= 2 &
  df$AwayTeamScore <= 3
] <- "2-3 goals"

df$AwayScoreGroup[df$AwayTeamScore >= 4] <- "4 or more goals"

# Convert the new variable to a factor.
df$AwayScoreGroup <- as.factor(df$AwayScoreGroup)

# Check the number of matches in each group.
table(df$AwayScoreGroup)

```

```

##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26

```

```

# Count the number of matches in each group.
away_group_counts <- table(df$AwayScoreGroup)

# Put the groups in order from the smallest count to the largest count.
away_group_order <- names(sort(away_group_counts))

# Give a light colour to the smallest group and a dark colour to the largest group.
away_group_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  away_group_order
)

# Calculate the mean and median HomeTeamScore for each group.
away_group_mean <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  mean,
  data = df
)

away_group_median <- aggregate(
  HomeTeamScore ~ AwayScoreGroup,
  median,
  data = df
)

away_group_mean

```

```

##      AwayScoreGroup HomeTeamScore

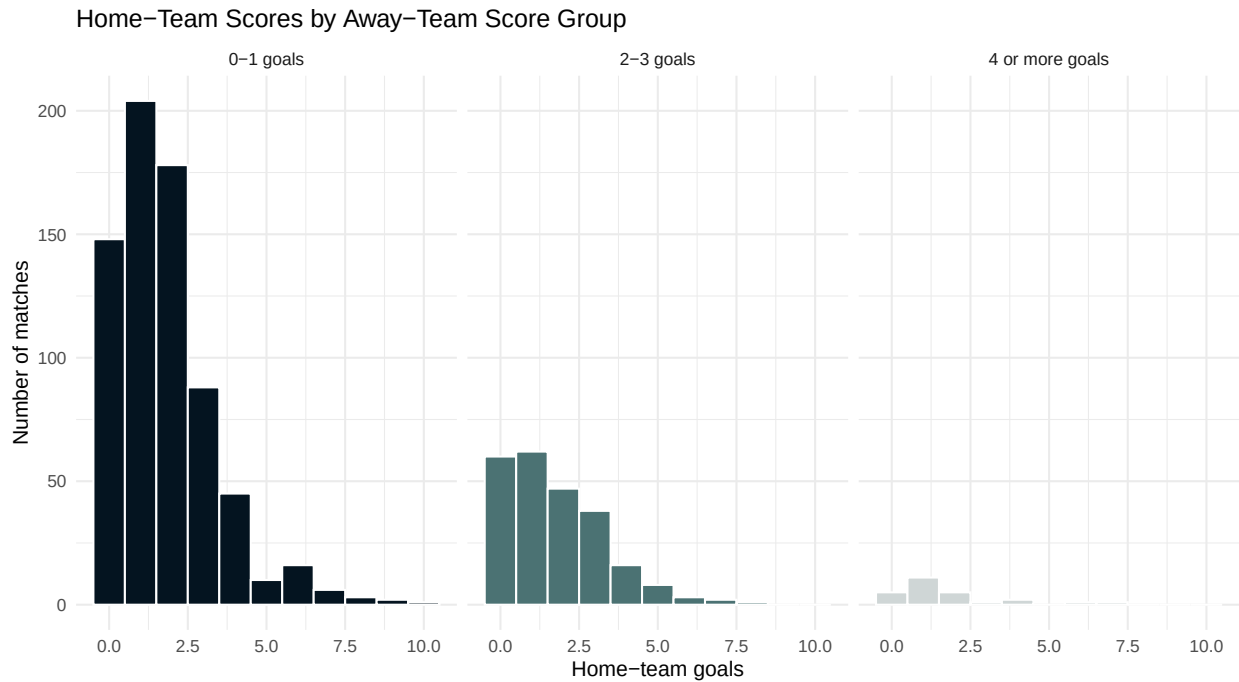
```

```
## 1      0-1 goals      1.774608
## 2      2-3 goals      1.746835
## 3 4 or more goals      1.730769
```

```
away_group_median
```

```
##      AwayScoreGroup HomeTeamScore
## 1      0-1 goals      1
## 2      2-3 goals      1
## 3 4 or more goals      1
```

```
ggplot(
  df,
  aes(
    x = HomeTeamScore,
    fill = AwayScoreGroup
  )
) +
  geom_histogram(
    binwidth = 1,
    colour = "white"
  ) +
  facet_wrap(~ AwayScoreGroup) +
  scale_fill_manual(
    values = away_group_colours
  ) +
  guides(fill = "none") +
  labs(
    title = "Home-Team Scores by Away-Team Score Group",
    x = "Home-team goals",
    y = "Number of matches"
  ) +
  theme_minimal()
```



Code Testing

```
# Check the score variable types.
class(df$HomeTeamScore)
```

```
## [1] "numeric"
```

```
class(df$AwayTeamScore)
```

```
## [1] "numeric"
```

```
# Check for missing score values.
sum(is.na(df$HomeTeamScore))
```

```
## [1] 0
```

```
sum(is.na(df$AwayTeamScore))
```

```
## [1] 0
```

```
# Check the new group variable.
class(df$AwayScoreGroup)
```

```
## [1] "factor"
```

```
table(df$AwayScoreGroup)
```

```
##
##      0-1 goals      2-3 goals 4 or more goals
##           701           237           26
```

```
sum(is.na(df$AwayScoreGroup))
```

```
## [1] 0
```

```
# Check that all matches are included in the three groups.
sum(table(df$AwayScoreGroup))
```

```
## [1] 964
```

```
nrow(df)
```

```
## [1] 964
```

Both the `HomeTeamScore` and the `AwayTeamScore` fields should be considered as numeric fields, and it is okay for R to show them as `"integer"`, because football scores are whole numbers.

The missing values should equal zero while the `AwayScoreGroup` variable should be factor and should not have any missing values.

The total number of matches in all the three groups should equal the amount of rows in the `df` data frame. Through this check it can be confirmed that every match has been put into one of the groups.

Results and Interpretation

The highest average of home-team scores was in final stage with an average of 3.83 goals, while the lowest home-team average appeared in the second group stage with a mean of 1.56 goals.

The histograms indicate that most home-team scores were located in the lower scores range, and that some stages had larger number of matches than others, making their bars taller. This chart can help understand the overall scoring process but in this case the mean is important as well since the number of matches was different.

The tournament with the highest home-team average score is the 1954 FIFA World Cup with 4.19 goals. The lowest scoring tournament is the 2010 FIFA World Cup with 1.19 goals.

The score distribution was not the same for all tournaments, since some tournaments had more scores but some tournaments had many games with zero, one or two goals. Nevertheless, as the number of matches was different in all tournament it is important to compare not only the graphs but also the means and medians of different tournaments.

In terms of away teams' score groups, 0-1 scored best on home teams' average with 1.77 goals. It means that the home teams' score differs among three away teams' scores groups, however this is only correlation because there is no causal effect.

Question 3: Penalty Shootout Participation

Rationale

I filtered the games where `PenaltyShootout` equals one, as Question 3 is limited to the games with a penalty shootout.

After that, I grouped the data by team name and team code and calculated the number of matches for each team in two steps. For this, I used `summarise()` and `n()`.

I ran the home team and away team analysis separately, since a team can be home in one game and away in another. Then I sorted the data by the number of matches in descending order and selected the first five teams through `head()`.

Filter Penalty Shootout Matches

```
# Keep only matches that involved a penalty shootout.
penalty_matches <- df %>%
  filter(PenaltyShootout == 1)

# Check the number of penalty-shootout matches.
nrow(penalty_matches)
```

```
## [1] 35
```

```
# Check that the filtered data contains only value 1.
table(penalty_matches$PenaltyShootout)
```

```
##
##  1
## 35
```

Top Five Home Teams

```
# Group the penalty matches by home-team name and code.
# Count the number of matches for each home team.
home_penalty_count <- penalty_matches %>%
  group_by(HomeTeamName, HomeTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
home_penalty_count <- home_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_home <- head(home_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_home) <- c(
```

```

  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five home teams as a table.
top5_home

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>           <int>
## 1 Argentina     ARG                 4
## 2 Brazil        BRA                 4
## 3 Spain         ESP                 4
## 4 Netherlands   NLD                 3
## 5 West Germany  DEU                 3

```

Top Five Away Teams

```

# Group the penalty matches by away-team name and code.
# Count the number of matches for each away team.
away_penalty_count <- penalty_matches %>%
  group_by(AwayTeamName, AwayTeamCode) %>%
  summarise(NumberOfMatches = n())

# Sort the teams from the highest count to the lowest count.
away_penalty_count <- away_penalty_count %>%
  arrange(desc(NumberOfMatches))

# Select the first five teams.
top5_away <- head(away_penalty_count, 5)

# Change the column names to match the assignment requirement.
names(top5_away) <- c(
  "TeamName",
  "TeamCode",
  "NumberOfMatches"
)

# Display the top five away teams as a table.
top5_away

```

```

## # A tibble: 5 x 3
## # Groups:   TeamName [5]
##   TeamName      TeamCode NumberOfMatches
##   <chr>         <chr>           <int>
## 1 France       FRA                 5
## 2 Argentina    ARG                 3
## 3 England      ENG                 3
## 4 Croatia      HRV                 2
## 5 Italy         ITA                 2

```

Code Testing

```
# Check that all filtered matches involved a penalty shootout.  
table(penalty_matches$PenaltyShootout)
```

```
##  
## 1  
## 35
```

```
# Both top-five tables should contain five rows.  
nrow(top5_home)
```

```
## [1] 5
```

```
nrow(top5_away)
```

```
## [1] 5
```

```
# Check that both tables have the required column names.  
names(top5_home)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
names(top5_away)
```

```
## [1] "TeamName"      "TeamCode"      "NumberOfMatches"
```

```
# Check for missing values in team names and codes.  
sum(is.na(top5_home$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_home$TeamCode))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamName))
```

```
## [1] 0
```

```
sum(is.na(top5_away$TeamCode))
```

```
## [1] 0
```

```
# Check that the match counts are sorted from high to low.
top5_home$NumberOfMatches
```

```
## [1] 4 4 4 3 3
```

```
top5_away$NumberOfMatches
```

```
## [1] 5 3 3 2 2
```

The `PenaltyShootout` table is expected to show the value of 1 only. This implies that the data filtered shows only penalty shootout matches.

The two final tables should consist of five rows each. The columns `TeamName`, `TeamCode` and `NumberOfMatches` must also be included. The checks for missing values should return zero in both cases. The count of matches should be listed from the highest value downwards.

Results and Interpretation

Argentina is the first team to be placed on the home table. This team participated in 4 penalty shootouts as the team that played at home.

France is the first team to be placed on the away table. This team participated in 5 penalty shootouts as the away team.

There are 1 teams that appeared in both teams. These teams were Argentina. The ranking of the teams based on the two tables is different based on the names of the teams and their order.

The order of the home and away tables can be different in some matches. In the case of this dataset, being the home team does not always mean that the team plays in its country.

Question 4: Factors Influencing Match Outcomes

Selected Factors and Rationale

The two factors that I have selected to work on for my investigation are **Stage** and **ExtraTime**. The rationale behind the selection of **Stage** is that matches can be played in different ways at different stages of the tournament. A match at a group stage cannot be the same as a match played during the semi-finals or finals. Different strategy might be used by the teams depending on the importance of the match.

The reason behind the selection of the **ExtraTime** variable is that there might be a different pattern of results in matches that continued into extra time. For instance, extra time might be played in matches when the teams are very close in score.

To investigate these variables, I first applied frequency tables to compute the frequency of home wins, away wins or draws. Then I made use of row percentages because the number of the matches in different groups was not equal. Proportionally constructed charts were used in order to visualize the comparison of the three result categories.

Overall Match Results

```
# Count the number of matches in each Result category.
overall_result_table <- table(df$Result)
```

```
overall_result_table
```

```
##
## away team win      draw home team win
##           240           179           545
```

```
# Put the Result categories in order
# from the smallest count to the largest count.
result_order <- names(sort(overall_result_table))
```

```
# Give a light colour to the least common result
# and a dark colour to the most common result.
```

```
result_colours <- setNames(
  c(
    lightest_colour,
    middle_colour,
    darkest_colour
  ),
  result_order
)
```

```
# Find the most common match result.
most_common_result <- names(overall_result_table)[
  which.max(overall_result_table)
]
```

```
most_common_result_count <- max(overall_result_table)
```

Factor 1: Competition Stage

```
# Create a table of Stage and Result.
```

```
stage_result_count <- table(  
  df$Stage,  
  df$Result  
)
```

```
stage_result_count
```

```
##  
##               away team win draw home team win  
##   final               4     0               17  
##   final round          0     1               5  
##   group stage        164   166             346  
##   quarter-finals      24     2             44  
##   round of 16         21     2             74  
##   second group stage  11     8             17  
##   semi-finals        10     0             28  
##   third-place match   6     0             14
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each Stage.
```

```
stage_result_percent <- prop.table(  
  stage_result_count,  
  margin = 1  
) * 100
```

```
# Round the percentages to one decimal place.
```

```
stage_result_percent <- round(  
  stage_result_percent,  
  1  
)
```

```
stage_result_percent
```

```
##  
##               away team win draw home team win  
##   final               19.0  0.0             81.0  
##   final round          0.0 16.7             83.3  
##   group stage        24.3 24.6             51.2  
##   quarter-finals      34.3  2.9             62.9  
##   round of 16         21.6  2.1             76.3  
##   second group stage  30.6 22.2             47.2  
##   semi-finals        26.3  0.0             73.7  
##   third-place match   30.0  0.0             70.0
```

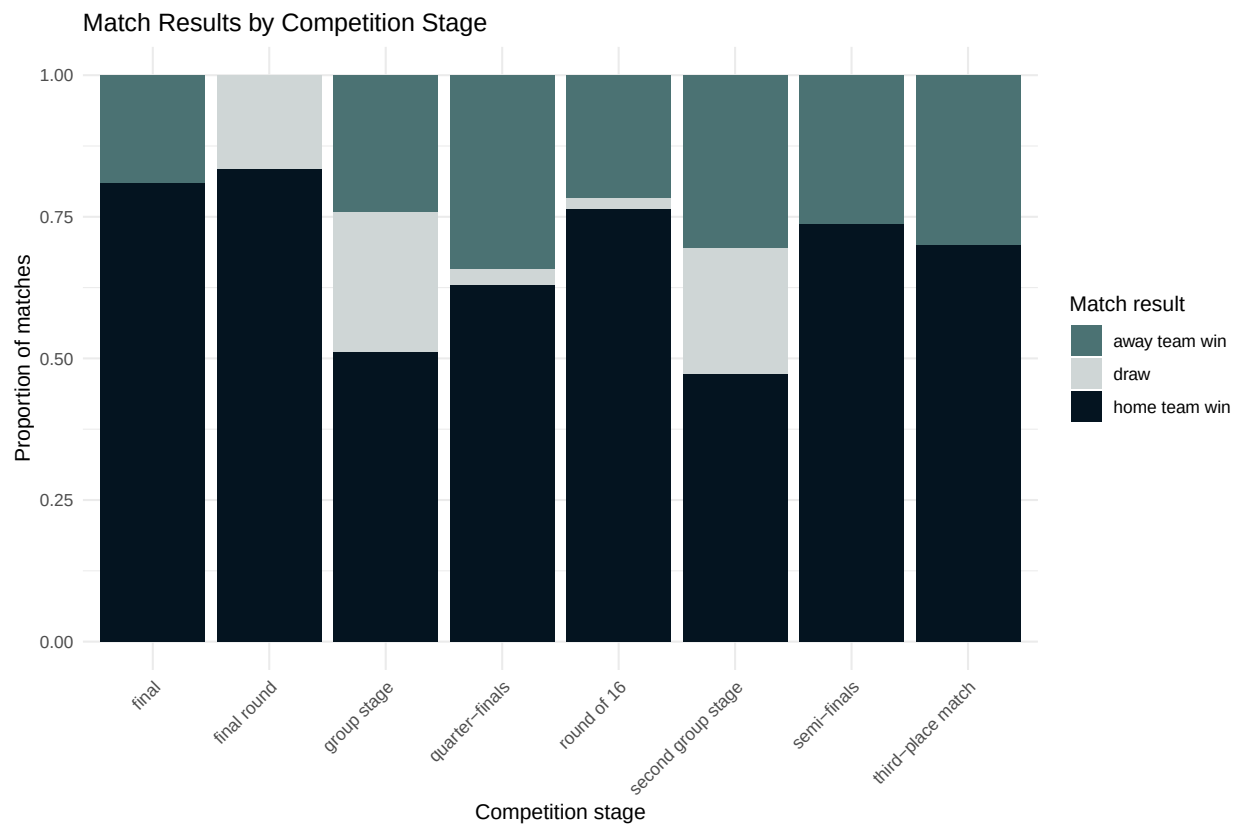
```
# Create a proportional bar chart of Result for each Stage.
```

```
ggplot(  
  df,  
  aes(  
    x = Stage,
```

```

    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Competition Stage",
    x = "Competition stage",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 45,
      hjust = 1
    )
  )
)

```



Factor 2: Extra Time

```
# Create a table of ExtraTime and Result.
```

```
extra_time_result_count <- table(
  df$ExtraTime,
  df$Result
)
```

```
extra_time_result_count
```

```
##
```

```
##      away team win draw home team win
```

```
##    0          215  173          503
```

```
##    1           25   6           42
```

```
# Convert the counts into row percentages.
```

```
# margin = 1 calculates percentages within each ExtraTime group.
```

```
extra_time_result_percent <- prop.table(
  extra_time_result_count,
  margin = 1
) * 100
```

```
# Round the percentages to one decimal place.
```

```
extra_time_result_percent <- round(
  extra_time_result_percent,
  1
)
```

```
extra_time_result_percent
```

```
##
```

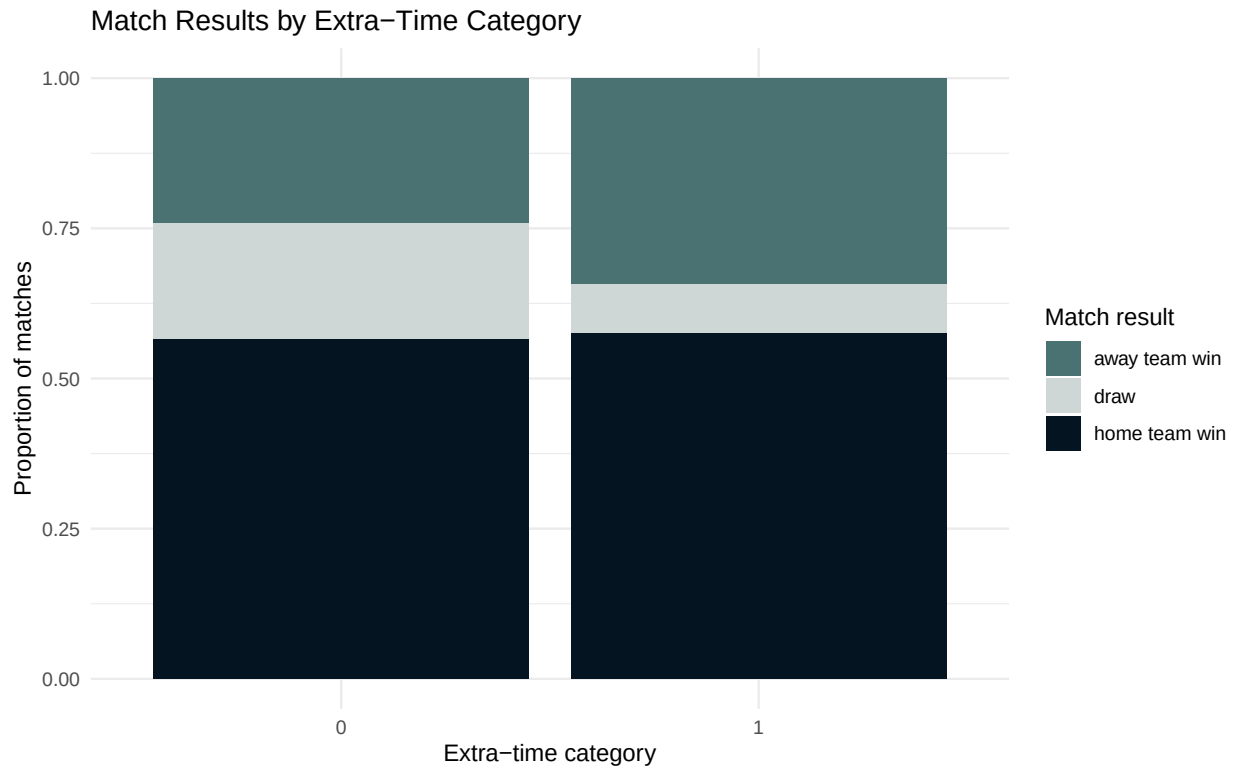
```
##      away team win draw home team win
```

```
##    0          24.1 19.4          56.5
```

```
##    1          34.2  8.2          57.5
```

```
# Create a proportional bar chart of Result by ExtraTime.
```

```
ggplot(
  df,
  aes(
    x = ExtraTime,
    fill = Result
  )
) +
  geom_bar(position = "fill") +
  scale_fill_manual(
    values = result_colours
  ) +
  labs(
    title = "Match Results by Extra-Time Category",
    x = "Extra-time category",
    y = "Proportion of matches",
    fill = "Match result"
  ) +
  theme_minimal()
```

Code Testing

```
# Check that the three variables are factors.
class(df$Result)
```

```
## [1] "factor"
```

```
class(df$Stage)
```

```
## [1] "factor"
```

```
class(df$ExtraTime)
```

```
## [1] "factor"
```

```
# Display the categories in each variable.
levels(df$Result)
```

```
## [1] "away team win" "draw"          "home team win"
```

```
levels(df$Stage)
```

```
## [1] "final"          "final round"    "group stage"
## [4] "quarter-finals" "round of 16"    "second group stage"
## [7] "semi-finals"    "third-place match"
```

```

levels(df$ExtraTime)

## [1] "0" "1"

# Check for missing values.
sum(is.na(df$Result))

## [1] 0

sum(is.na(df$Stage))

## [1] 0

sum(is.na(df$ExtraTime))

## [1] 0

# The total number in each table should equal the number of rows in df.
sum(stage_result_count)

## [1] 964

nrow(df)

## [1] 964

sum(extra_time_result_count)

## [1] 964

nrow(df)

## [1] 964

# Each row of the percentage tables should total approximately 100%.
rowSums(stage_result_percent)

##           final      final round      group stage      quarter-finals
##           100.0          100.0          100.1          100.1
## round of 16 second group stage      semi-finals third-place match
##           100.0          100.0          100.0          100.0

rowSums(extra_time_result_percent)

##      0      1
## 100.0  99.9

```

The outputs for `Result`, `Stage`, and `ExtraTime` are required to be "factor" since they demonstrate the various groups of categories present inside each variable respectively. The results from the missing-values test must be zero. Each of the counts must yield the same output as the rows in `df`. The percentage tables must add up to 100% too, although results like 99.9% or 100.1% would not be uncommon since the percentages were rounded up.

Results and Interpretation

In all matches, the home team winning was the most frequently observed result and the total number of matches resulting in this outcome was 545.

Considering the stages of the competition, the last stage had the most percentage of home team winning, which is equal to 83.3%. The table of percentages indicates that the same trend is not observed for every competition stage.

Some stages show a higher ratio of observed results; while in a different part of stages the losses of the home team or draws were more frequent.

As for the extra time part, the first category showed the highest ratio of home team winning with 57.5%. Moreover, the table demonstrates that the distribution of the results differs for the extra time categories.

Thus, it can be concluded that the stages of the competition and extra time play a role in the match result in the data set discussed. However, in the analysis of the obtained results it was not proven whether the impact of both mentioned factors was responsible for home team win, home team loss, or draw.

Question 5: Git and GitHub

GitHub Repository

I created and managed the submission repository by using Git commands in Ubuntu Terminal.

The repository is available at:

<https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission>

The analysis for Questions 1 to 4 was already prepared before this repository was created. I then organised the completed files into a new project folder and used Terminal commands to create the repository, make separate commits and push the project to GitHub.

Project Folder Structure

The project was organised into separate folders:

- **data/** contains the four FIFA World Cup datasets.
- **scripts/** contains the R scripts for Questions 1 to 4.
- **report/** contains the R Markdown file and the final PDF report.
- **figures/** contains the Terminal screenshots used as evidence.
- **COMP1013-FIFA-WorldCup-Analysis.Rproj** is the RStudio project file.

The folders keep the datasets, scripts, report and figures separate. This makes the project easier to understand and check.

Terminal Git Workflow Evidence

Preparing the New Project

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ pwd
/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la
ls data
ls scripts
ls report
ls figures
total 4
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 .
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:56 ..
-rwxrwxrwx 1 yvette yvette 719 Jul 20 02:57 .gitignore
-rwxrwxrwx 1 yvette yvette 218 Jul 20 02:57 COMP1013-FIFA-WorldCup-Analysis.Rproj
-rwxrwxrwx 1 yvette yvette 139 Jul 20 02:57 README.md
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 data
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 figures
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 report
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 scripts
Matches.csv Stadiums.csv Teams.csv Tournaments.csv
01_data_preparation.R 03_penalty_shootouts.R
02_score_distribution.R 04_match_outcomes.R
Report.Rmd Report.pdf
commit_history.png
```

This screenshot shows a new submission directory as well as its project structure. Datasets, scripts, report and figures were allocated to particular folders before introducing the Git system.

Initialising Git

```
Initialized empty Git repository in /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission/.git/
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git branch -M main
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        COMP1013-FIFA-WorldCup-Analysis.Rproj
        README.md
        data/
        figures/
        report/
        scripts/

nothing added to commit but untracked files present (use "git add" to track
)
```

I made use of `git init` in order to set up the new local repository. In addition, I renamed the main branch to `main`. `git status` results indicate that the repository does not have any commits yet and no files are accounted for.

First Commit: Project Configuration

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add .gitignore
git add README.md
git add COMP1013-FIFA-WorldCup-Analysis.Rproj
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   .gitignore
    new file:   COMP1013-FIFA-WorldCup-Analysis.Rproj
    new file:   README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    data/
    figures/
    report/
    scripts/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
.gitignore | 52 ++++++
COMP1013-FIFA-WorldCup-Analysis.Rproj | 13 +++++
README.md | 2 +
3 files changed, 67 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Set up project"
[main (root-commit) a25e784] Set up project
3 files changed, 67 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 COMP1013-FIFA-WorldCup-Analysis.Rproj
 create mode 100644 README.md
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
a25e784 (HEAD -> main) Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    data/
    figures/
    report/
```

I applied `git add` to stage the project configuration files and successfully made the first commit using `git commit` by coding the message `Set up project`.

Adding the Datasets

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la data
total 128
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57 
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01 
-rwxrwxrwx 1 yvette yvette 107135 Jul 20 02:57 Matches.csv
-rwxrwxrwx 1 yvette yvette 8299 Jul 20 02:57 Stadiums.csv
-rwxrwxrwx 1 yvette yvette 1595 Jul 20 02:57 Teams.csv
-rwxrwxrwx 1 yvette yvette 696 Jul 20 02:57 Tournaments.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add data/*.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   data/Matches.csv
        new file:   data/Stadiums.csv
        new file:   data/Teams.csv
        new file:   data/Tournaments.csv

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        figures/
        report/
        scripts/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
data/Matches.csv      | 965 +++++
data/Stadiums.csv     | 194 +++++
data/Teams.csv        | 86 +++++
data/Tournaments.csv  | 24 ++
4 files changed, 1269 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Add FIFA World Cup datasets"
[main a976ebb] Add FIFA World Cup datasets
4 files changed, 1269 insertions(+)
 create mode 100644 data/Matches.csv
 create mode 100644 data/Stadiums.csv
 create mode 100644 data/Teams.csv
 create mode 100644 data/Tournaments.csv
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
a976ebb (HEAD -> main) Add FIFA World Cup datasets
a25e784 Set up project
```

I uploaded the four CSV datasets one at a time. The commit message Add FIFA World Cup datasets recaps about the files uploaded.

Adding the R Scripts

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ ls -la scripts
total 16
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01
-rwxrwxrwx 1 yvette yvette 1950 Jul 20 02:57 01_data_preparation.R
-rwxrwxrwx 1 yvette yvette 2361 Jul 20 02:57 02_score_distribution.R
-rwxrwxrwx 1 yvette yvette 2010 Jul 20 02:57 03_penalty_shootouts.R
-rwxrwxrwx 1 yvette yvette 3132 Jul 20 02:57 04_match_outcomes.R
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git add scripts/*
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   scripts/01_data_preparation.R
    new file:   scripts/02_score_distribution.R
    new file:   scripts/03_penalty_shootouts.R
    new file:   scripts/04_match_outcomes.R

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git diff --cached --stat
scripts/01_data_preparation.R | 86 +++++
scripts/02_score_distribution.R | 113 +++++
scripts/03_penalty_shootouts.R | 80 +++++
scripts/04_match_outcomes.R | 136 +++++
4 files changed, 415 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git commit -m
"Add R scripts for Questions 1 to 4"
[main 10elc8a] Add R scripts for Questions 1 to 4
4 files changed, 415 insertions(+)
create mode 100644 scripts/01_data_preparation.R
create mode 100644 scripts/02_score_distribution.R
create mode 100644 scripts/03_penalty_shootouts.R
create mode 100644 scripts/04_match_outcomes.R
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --one
line --decorate
10elc8a (HEAD -> main) Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/

nothing added to commit but untracked files present (use "git add" to track)
```

I uploaded the four R scripts used to answer the Questions 1 through 4. The scripts were uploaded with the commit message Add R scripts for Questions 1 to 4.

Adding the R Markdown Report

```
yvette@dntvish: /mnt/c/User: × + ∨

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
ls -la report
total 368
drwxrwxrwx 1 yvette yvette 512 Jul 20 02:57
drwxrwxrwx 1 yvette yvette 512 Jul 20 03:01
-rwxrwxrwx 1 yvette yvette 29403 Jul 20 02:57 Report.Rmd
-rwxrwxrwx 1 yvette yvette 342000 Jul 20 02:57 Report.pdf
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git add report/Report.Rmd
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   report/Report.Rmd

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/Report.pdf

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git diff --cached --stat
report/Report.Rmd | 880 ++++++
1 file changed, 880 insertions(+)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git commit -m "Add R Markdown analysis report"
[main 86ae4f7] Add R Markdown analysis report
1 file changed, 880 insertions(+)
create mode 100644 report/Report.Rmd
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git log --oneline --decorate
86ae4f7 (HEAD -> main) Add R Markdown analysis report
10e1c8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Ana
lysis-Submission$ |
```

I uploaded `Report.Rmd` independently. Thus, the log contains the report source separately from data and R scripts.

Creating the GitHub Repository and Pushing

```
yvette@dntvish: /mnt/c/User: × + ▾
nothing added to commit but untracked files present (use "git add" to track
)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ gh auth status
github.com
✓ Logged in to github.com account yvette195 (/home/yvette/.config/gh/host
s.yml)
- Active account: true
- Git operations protocol: https
- Token: gho_*****
- Token scopes: 'gist', 'read:org', 'repo', 'workflow'
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ git remote -v
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ gh repo create COMP1013-FIFA-WorldCup-Analysis-Sub
mission \
  --public \
  --source=. \
  --remote=origin \
  --push
✓ Created repository yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission o
n github.com
https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission
✓ Added remote https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis
-Submission.git
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 8 threads
Compressing objects: 100% (22/22), done.
Writing objects: 100% (23/23), 34.31 KiB | 305.00 KiB/s, done.
Total 23 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.
git
* [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
✓ Pushed commits to https://github.com/yvette195/COMP1013-FIFA-WorldCup-Ana
lysis-Submission.git
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-World
Cup-Analysis-Submission$ |
```

I created the new public repository with the help of GitHub CLI in Ubuntu Terminal. The local repository was configured with GitHub as origin and the original commits were uploaded to the remote main branch.

Checking the GitHub Connection

```
yvette@dntvish: /mnt/c/Users/ yvette@
Total 23 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), done.
To https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
* [new branch]      HEAD -> main
branch 'main' set up to track 'origin/main'.
✓ Pushed commits to https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git remote -v

origin https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git (fetch)
origin https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git (push)
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git branch -vv
* main 86ae4f7 [origin/main] Add R Markdown analysis report
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --oneline --decorate -5
86ae4f7 (HEAD -> main, origin/main) Add R Markdown analysis report
10e1c8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    figures/
    report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish: /mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ |
```

The `git remote -v` command will show the new GitHub repository address. In comparison with local section, branch details point out that local `main` section follows `origin/main`. The log will confirm that both local and remote sections include the same commits.

Adding and Pushing the Terminal Evidence

```
yvette@dntvish: /mnt/c/User: × + ∨

nothing added to commit but untracked files present (use "git add" to track)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git add figures/*.png
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   figures/commit_history.png
    new file:   figures/terminal_01_prepare_new_project.png
    new file:   figures/terminal_02_git_init.png
    new file:   figures/terminal_03_first_commit.png
    new file:   figures/terminal_04_add_datasets.png
    new file:   figures/terminal_05_add_scripts.png
    new file:   figures/terminal_06_add_rmarkdown_report.png
    new file:   figures/terminal_07_create_repo_and_push.png
    new file:   figures/terminal_08_verify_github_connection.png

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    report/Report.pdf

yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git diff --cached --stat
figures/commit_history.png      | Bin 0 -> 90895 bytes
figures/terminal_01_prepare_new_project.png | Bin 0 -> 112459 bytes
figures/terminal_02_git_init.png | Bin 0 -> 60799 bytes
figures/terminal_03_first_commit.png | Bin 0 -> 127064 bytes
figures/terminal_04_add_datasets.png | Bin 0 -> 144668 bytes
figures/terminal_05_add_scripts.png | Bin 0 -> 157007 bytes
figures/terminal_06_add_rmarkdown_report.png | Bin 0 -> 161856 bytes
figures/terminal_07_create_repo_and_push.png | Bin 0 -> 162161 bytes
figures/terminal_08_verify_github_connection.png | Bin 0 -> 169469 bytes
9 files changed, 0 insertions(+), 0 deletions(-)
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git commit -m "Add Terminal Git workflow screenshots"
[main 033ad17] Add Terminal Git workflow screenshots
9 files changed, 0 insertions(+), 0 deletions(-)
create mode 100644 figures/commit_history.png
create mode 100644 figures/terminal_01_prepare_new_project.png
create mode 100644 figures/terminal_02_git_init.png
create mode 100644 figures/terminal_03_first_commit.png
create mode 100644 figures/terminal_04_add_datasets.png
create mode 100644 figures/terminal_05_add_scripts.png
create mode 100644 figures/terminal_06_add_rmarkdown_report.png
create mode 100644 figures/terminal_07_create_repo_and_push.png
create mode 100644 figures/terminal_08_verify_github_connection.png
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git pull --rebase origin main
git push origin main
From https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission
 * branch      main       -> FETCH_HEAD
Current branch main is up to date.
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 8 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.01 MiB | 7.25 MiB/s, done.
Total 12 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/yvette195/COMP1013-FIFA-WorldCup-Analysis-Submission.git
 86ae4f7..033ad17  main -> main
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git status
git log --oneline --decorate -6
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    report/Report.pdf

nothing added to commit but untracked files present (use "git add" to track)
033ad17 (HEAD -> main, origin/main) Add Terminal Git workflow screenshots
86ae4f7 Add R Markdown analysis report
10elc8a Add R scripts for Questions 1 to 4
a976ebb Add FIFA World Cup datasets
a25e784 Set up project
```

I uploaded the terminal commands with `git add` command and later made a commit with the message Add Terminal Git commands screenshots.

Commit History

```
yvette@dntvish:/mnt/c/Users/Admin/Documents/AP-Projects/COMP1013-FIFA-WorldCup-Analysis-Submission$ git log --oneline --decorate --graph -10
* 033ad17 (HEAD -> main, origin/main) Add Terminal Git workflow screenshots
* 86ae4f7 Add R Markdown analysis report
* 10e1c8a Add R scripts for Questions 1 to 4
* a976ebb Add FIFA World Cup datasets
* a25e784 Set up project
```

The diagram presents an account of the commits made to a new submission repository. Each of the commits was made one at a time for project configuration, datasets, R scripts, R Markdown reports, and terminal screenshots.

Explanation of the Commit History

According to the commit message log, I made all of the contributions to the project separately, not in a single commit.

1. **Set up project**

This commit involved the addition of the `.gitignore`, `README`, and RStudio project files.

2. **Add FIFA World Cup datasets**

This commit relates to the four CSV files shared in our assignment.

3. **Add R scripts for Questions 1 to 4**

The four R scripts added to prepare and analyze data are mentioned in this commit.

4. **Add R Markdown analysis report**

This commit incorporates the `Report.Rmd` file.

5. **Add Terminal Git workflow screenshots**

This commit brings in the screenshots showing the usage of Git and GitHub commands on Ubuntu Terminal.

Analyzing the commit logs, one can see what was done in different phases of developing the project. The messages regarding each commit are quite informative yet simple to understand.

The screenshot of this step was taken prior to the last commit since these images had to be placed in the report before the final output.